Satellite Radiance Data Assimilation Considering Spatiotemporal Variability of Ozone and Its Application in the Arctic

Xi Luo, Yaodeng Chen, Peiming Dong, Phinous O. Othoo, Deming Meng, Xusheng Yan, and Hong Zheng

***Abstract*—Ozone has a significant impact on radiance observations in specific bands of infrared and microwave sounders. However, current radiance simulations for assimilation oversimplify ozone's impact by using a single and invariant ozone profile, which neglects its spatiotemporall variability. This treatment degrades the accuracy of satellite radiance simulations in the corresponding bands. In order to mitigate the impact of this oversimplification, this study investigates a assimilation scheme for satellite radiance data that accounts for the spatiotemporal variability of ozone concentration. We conduct a two-month continuous assimilation and forecast experiment, which assimilates infrared and microwave observations from NOAA18-HIRS4, FY3C/D-MWHS-II, SNPP-ATMS, and MetOpC-MHS in the Arctic. It is demonstrated that the use of spatiotemporally varying ozone concentrations reduces biases in the simulation of the infrared and microwave ozone-sensitive bands, with brightness temperature simulation errors reduced by up to 5 K, and improves the variational bias correction (VarBC). This enhancement allows more observations to pass quality control and increases data availability (e.g., a 3.7% increase for HIRS4 9.71 μm). The assimilation experiments reveal that the combined assimilation of infrared and microwave observations using spatiotemporally varying ozone improves the analyses and forecasts, reducing temperature and humidity errors by 1–6 K and 0.5%–1%, leading to a more accurate simulation of the Arctic atmosphere.**

***Index Terms*—Arctic, ozone, satellite radiance data assimilation.**

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I. Introduction

SATELLITE radiance data assimilation has significantly enhanced global and regional numerical forecasting capabilities [1], particularly in the oceans and polar areas, where conventional observations are limited. At ECMWF, over 90% of assimilated observations within the Arctic are satellite infrared and microwave radiance data [2].

The rapid radiative transfer model (RRTM) is the crucial observation operator for direct assimilation of satellite radiance data. It constructs a link between atmospheric state variables and satellite radiance measurements, simulating radiance at the top of the atmosphere using vertical profiles of temperature, water vapor mixing ratio, ozone, as well as surface variables, geographic information, and instrument parameters [3], [4]. Ozone, one of the most important trace gases in the atmosphere, has a substantial influence on radiance brightness temperature calculations in specific bands.

During assimilation, atmospheric state variables and some other parameters required for RRTM calculations are mostly provided by the background or observations. However, trace gases, such as ozone, are often simplified to the default profile provided by the RRTM in most operational data assimilation systems [5]. This simplification is not in line with the spatiotemporally varying characteristics of ozone in the actual atmospheric, especially in the Arctic region.

It has been shown that ozone has a significant effect on radiances within the ozone-sensitive bands of infrared sounders such as IASI [6], [7]. The simulated radiance using more realistic ozone data is closer to observations [8]. Ozone simulations from the chemical model are implemented to RTTOV in Météo-France's ARPEGE 4-D-variational global modeling system to assimilate ozone-sensitive observations from infrared hyperspectral sounders such as IASI. This implementation improves the analysis of temperature, relative humidity, and wind, as well as the subsequent forecast [4]. Emma et al. [9] demonstrated that the impact of ozone on the new generation of European microwave sensor Ice Cloud Imager (ICI), which operates at frequencies up to 664 GHz, is nonnegligible. In fact, ozone also has an impact on the existing 183 GHz bands. Line-by-line radiative transfer model shows that omitting ozone in 183 GHz can result in brightness temperature simulation biases of up to 0.5 K [10], [11].

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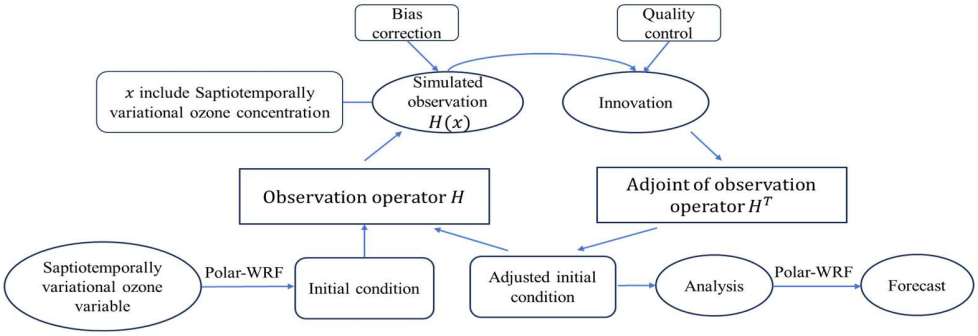


Fig. 1. Implementation process of the assimilation scheme considering the spatiotemporal variability of ozone.

However, the full potential impact of ozone characteristics on assimilations and forecasts requires further in-depth work.

In the Arctic region, where conventional observations are scarce and the accuracy of numerical forecasts depends highly on satellite observations, the spatiotemporal variability of ozone is more pronounced than in other regions [12]. Thus, the treatment of trace gases in the RRTM requires more careful consideration [13]. To improve satellite radiance data assimilation in the Arctic, this study develops a joint assimilation scheme of infrared and microwave radiance observations considering the spatiotemporal variability of ozone. This scheme is applied to the satellite observations from both the IR sounder HIRS4 flown onboard NOAA18 and the microwave instruments onboard FY3C/D, SNPP and MetOpC.

This article is organized as follows. Sections II and III describe the assimilation scheme, the model set-up, and experimental design, respectively. Section IV presents the sensitivity analysis of brightness temperature simulations to ozone in the Arctic. Section V discusses the assimilation and forecast experiments based on the spatiotemporal variability of ozone. The conclusion and discussion are given in Section VI.

II. Method

The assimilation scheme considering the spatiotemporal variability of ozone is carried out in the WRFDA 3-D variational module. The 3-D variational assimilation solves a quadratic minimization problem for the deviation of the analysis from the background and observation with a cost function

$$J(x) = \frac{1}{2}(x - x_b)^{\mathrm{T}} B^{-1}(x - x_b) + \frac{1}{2}(H(x) - y_o)^{\mathrm{T}} \times R^{-1}(H(x) - y_o) \quad (1)$$

where x is the analysis field, x_b is the background field and y_o stands for the observation; B and R are the background and observation error covariance matrices, respectively; H, which is the observation operator, commonly uses the RRTM for satellite radiance observations.

The RRTM used in this study is RTTOV13. This version has been significantly updated from the previous versions. It not only supports microwave radiance simulations with ozone profiles as input predictors, but also provides new optical thickness coefficients accounting for variable ozone [14]. These enhancements provide technical support for considering the spatiotemporal variability of ozone when assimilating microwave radiance observations.

Within the assimilation framework, $H(x)$ transforms model state variables into observation space equivalents. Here x includes temperature, water vapor mixing ratio, surface temperature, and other variables such as ozone. Our scheme replaces the default single ozone profile of RRTM with the spatiotemporally varying ozone profiles. The operator is reformulated as $H[x(\text{lat, lon, level, } t)]$ instead of $H[x(\text{level})]$ to better represent realistic atmospheric conditions. The implementation process of the scheme is illustrated in Fig. 1.

The implementation involves three key steps.

1) Either reanalysis or forecasts are used to provide spatiotemporally varying ozone concentrations for the background.
2) In the observation operator, spatiotemporally varying ozone profiles are used in its forward mode calculation to simulate satellite radiance brightness temperature. It is followed by bias correction and quality control procedures to obtain the updated innovation vector.
3) In the assimilation process, both tangent-linear and adjoint mode calculations of RTTOV13 are carried out with spatiotemporally varying ozone concentration, so as to iteratively obtain the optimal atmospheric analysis at that moment, and for subsequent assimilations and forecasts.

III. Model Set-Up

The forecast model for this experiment is Polar-WRF version 4.3.3, a polar-optimized version of the Weather Research and Forecasting (WRF) model developed by the Polar Meteorology Group at the Byrd Polar and Climate Research Center of Ohio State University [15]. The unique environmental characteristics of the polar regions are taken into account by optimizing the heat transport scheme [15], [16], improving surface energy balance in the land surface model [17], and enhancing the cloud microphysics parameterization scheme for the Arctic region [18].

The model uses a single domain covering all areas north of 60°N at 15-km horizontal resolution centered on the North Pole with a model top at 5 hPa. The physical parameterization scheme for this experiment is as follows: aerosol-aware Thompson microphysics scheme [19], RRTMG longwave and shortwave radiation scheme [20], the MYNN2 boundary layer

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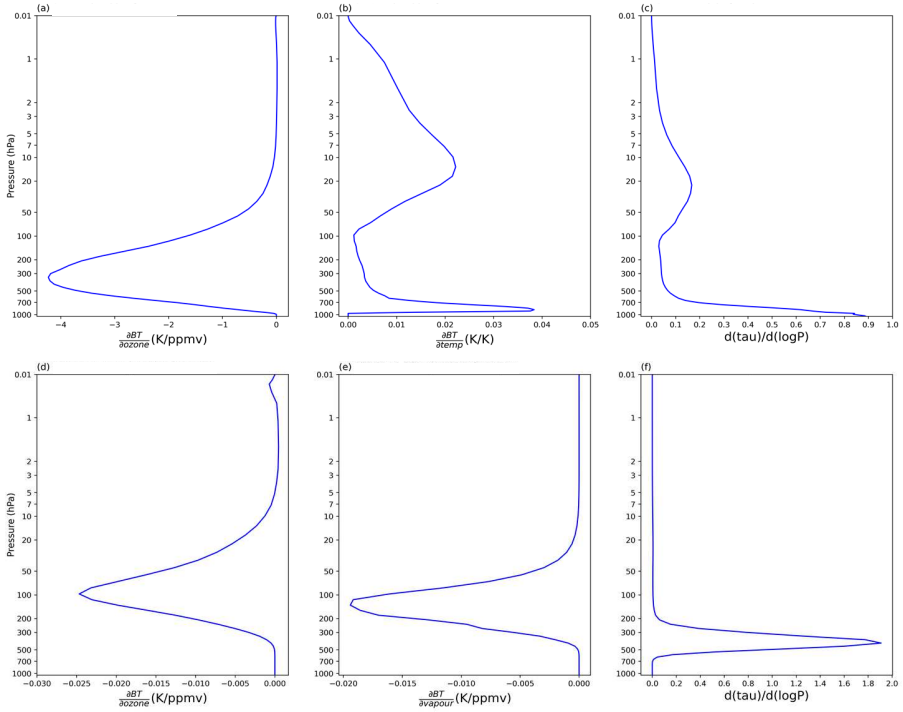


Fig. 2. Jacobians and weighting functions for (a)~(c) HIRS4 9.71 μm and (d)~(f) 183.31 $\pm$ 1.0 GHz band.

scheme [21], the Noah-MP land-surface model scheme [22], and the Kain–Fritsch cumulus scheme [23].

Two parallel experiments are conducted to evaluate the effect of using ozone profiles with different characteristics on the assimilation. One experiment (EXP_default) uses a single default ozone profile provided by RTTOV13, while the other (EXP_ozone) uses real-time ozone profiles provided by ERA5 reanalysis. Both experiments assimilate identical observations: the 9.71 μm band of NOAA18-HIRS4 and the 183.31 $\pm$ 1.0 GHz band of FY3C/D-MWHS-II, SNPP-ATMS, and MetOpC-MHS. The 9.71 μm band from HIRS4 mainly observes the temperature and ozone concentration in the altitude range of 100–5 hPa [see Fig. 2(a) and (b)]. The brightness temperature of this band shows a negative sensitivity to ozone concentration from 500 to 50 hPa, meaning that higher ozone concentration will result in lower brightness temperature [see Fig. 2(a)]. Given the difficulty in interpreting the assimilation impact of this band on its sensitivity to the complex Arctic surface, and as this topic is beyond the study's scope, its effects on lower-level analysis and forecasts are excluded from the subsequent discussion. The 183.31 $\pm$ 1.0 GHz band of the MWHS-II, ATMS, and MHS can mainly observe the atmospheric humidity information in the altitude range of 500–300 hPa [see Fig. 2(e)], and the brightness temperature of this band exhibits negative sensitivity to ozone concentration in the altitude range of 300–10 hPa, which is slightly less affected by ozone compared to the infrared band [see Fig. 2(d)].

The two-month assimilation and forecast experiment from November 1 to December 31, 2023, is conducted in this study. A 6-h spin-up is performed daily at 0000UTC, followed by assimilation every 6 h, with the 6-h forecast from the previous moment as the background. The assimilation cycle continues until 0000 UTC the following day, at which point a 24-h forecast was made. These steps are repeated daily. Clear-sky assimilation is performed for observations from all satellite instruments. Radiance data must undergo quality control before assimilation to eliminate observations with large errors that could negatively impact the assimilation process. The quality control scheme for HIRS4 9.71 μm in this experiment is as follows.

1) All observations over mixed surfaces, land, sea ice, and snow-covered surfaces are removed.
2) Observations at limb positions are excluded.
3) Observations with cloud liquid water path (CLWP) $\geq$ 0.2 mm are excluded.
4) Observations where the absolute OMB value exceeds 15 K or three times the observation error are discarded.

The quality control scheme for microwave 183.31 $\pm$ 1.0 GHz is as follows.

1) Because of challenges in calculating surface emissivity over the complex Arctic terrain, all observations over mixed surfaces are removed.
2) Observations at limb positions are excluded.
3) Observations with CLWP $\geq$ 0.2 mm are excluded. For MWHS-II, a scattering index (SI) is defined as SI = TB(1) – TB(10), representing the brightness temperature difference between Channel 1 and Channel 10. Observations are excluded if SI > 4 K.
4) Observations where the absolute OMB value exceeds 15 K or three times the observation error are discarded.

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IV. Sensitivity Analysis of Brightness Temperatures Simulations to Ozone

A. Spatiotemporal Characteristics of Ozone in the Arctic

To investigate the impact of ozone profiles with different characteristics on data assimilation, this study first analyzes the spatiotemporal characteristics of ozone in the Arctic and the discrepancies between actual ozone concentrations and the RTTOV default profile. Fig. 3 illustrates the monthly variations of spatially averaged ERA5 ozone concentrations focusing on ozone-sensitive altitudes (300 and 100 hPa) relevant to the HIRS4 9.71 μm band and microwave 183.31 $\pm$ 1.0 GHz band within the Arctic region [see Fig. 3(a)], as well as the hourly variations of ozone concentrations on December 1, 2023 [see Fig. 3(b)]. It can be seen that ozone in the Arctic region not only has significant monthly variations but also pronounced diurnal variability. Furthermore, the spatial distribution of monthly averaged ozone concentrations in December at 300 and 100 hPa [see Fig. 3(c) and (d)] also demonstrate notable regional variability. Therefore, during actual assimilation in the Arctic, ozone concentrations exhibit obvious spatiotemporal variability across different observation times and locations.

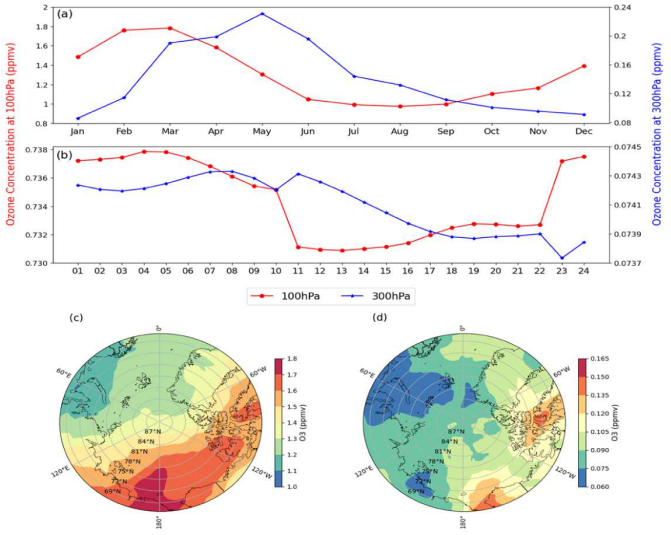


Fig. 3. Characteristics of (a) monthly variation of ERA5 ozone concentrations (ppmv) spatially averaged in the Arctic over 100 and 300 hPa during 2023, (b) hourly variation on 1 December, and the spatial distribution of monthly averaged ozone concentration in December 2023 at (c) 100 hPa and (d) 300 hPa.

The mean and standard deviation profiles of the difference between the ERA5 monthly mean ozone concentration and the default single ozone profile of RTTOV within the Arctic region are given in Fig. 4. It is revealed that the difference between the two profiles is mainly in the levels above 300 hPa. Between 300 and 30 hPa, the default profile consistently underestimates true ozone concentrations, with the maximum deviation (1.5 ppmv) occurring near 50 hPa. Conversely, from 30 to 3 hPa, the default profile significantly overestimates ozone values, peaking at 10 hPa where differences reach nearly 4 ppmv in certain months. It is notable that these differences persist throughout all months in the ozone-sensitive altitude range relevant to both HIRS4 9.71 μm and microwave 183.31 $\pm$ 1.0 GHz bands.

B. Impact of Ozone Profile Selection on Radiance Simulations

Preceding results have demonstrated significant temporal and spatio variability of ozone in the Arctic, along with notable differences between the default ozone profile of RTTOV and actual values. This section will further discuss the effects of these different ozone profiles on satellite radiance brightness temperature simulations.

Passive satellite observation is the portion of radiation that originates from the Earth's surface, travels through the atmosphere, and is finally received by the satellite. During the transmission, the radiation is influenced by atmospheric absorption and the atmosphere's own emission. The stronger the atmospheric absorption, the less radiation is received. This effect varies across different wavelength bands and manifests as differences in satellite channels. Ozone is one of the most important trace gases in the atmosphere. The Jacobian of satellite channel radiance with respect to ozone concentration represents a measurement of sensitivity. The higher the sensitivity, the greater the impact. Consequently, for satellite channels sensitive to ozone, the satellite observation is significantly affected by the ozone radiation effect. Using a fixed ozone profile in radiative transfer modeling has inherent limitations: when a fast radiative transfer model employs a fixed, globally averaged ozone profile, it effectively assumes that the radiative influence of ozone at each altitude remains constant worldwide and throughout the year, being a significant departure from reality. By using spatiotemporally variable ozone that is close to real atmospheric conditions, the accuracy of ozone radiation effect can be improved in radiative transfer, making the satellite simulations closer to the observations.

Figs. 5 and 6 present the observation-minus-background (OMB) mean and root-mean-square error (RMSE) in the Arctic region (north of 60°N) from 0600 UTC on November 1, to 0000 UTC on December 31, 2023, covering 9.71 μm band of NOAA18-HIRS4 and 183.31 $\pm$ 1.0 GHz bands of FY3C/D-MWHS-II, SNPP-ATMS, and MetOpC-MHS. For the 9.71 μm band, which is the most sensitive band to ozone, the use of spatiotemporally varying ozone profiles significantly reduces the negative bias of the simulations using the default profiles at all moments, with OMB mean and RMSE reductions reaching up to 5 K. For the simulations of microwave 183.31 $\pm$ 1.0 GHz bands, OMB mean and RMSE reductions are both around 0.15 K for MWHS-II, 0.5 and 0.1 K for MHS, respectively. The simulation errors of ATMS are reduced in the latter part of the experiment. Consequently, the use of spatiotemporally varying ozone profiles instead of the default profile can reduce the brightness temperature simulation bias in specific bands of the infrared and microwave, and make the simulation results closer to the observations in the Arctic.

V. Results of Assimilation and Forecast Experiments

A. Bias Correction

Variational bias correction (VarBC) serves to reduce systematic errors in satellite observations through adaptive adjustment

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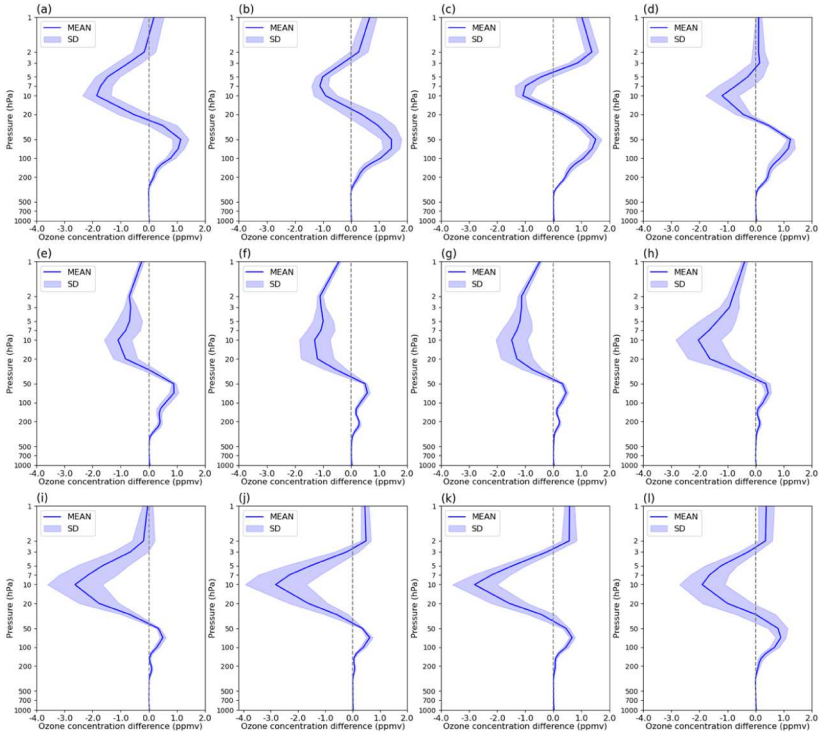


Fig. 4. Mean and standard deviation profiles of the difference between ERA5 ozone concentration and RTTOV default ozone profile (ERA5-default) in the Arctic, (a)–(l) January–December 2023.

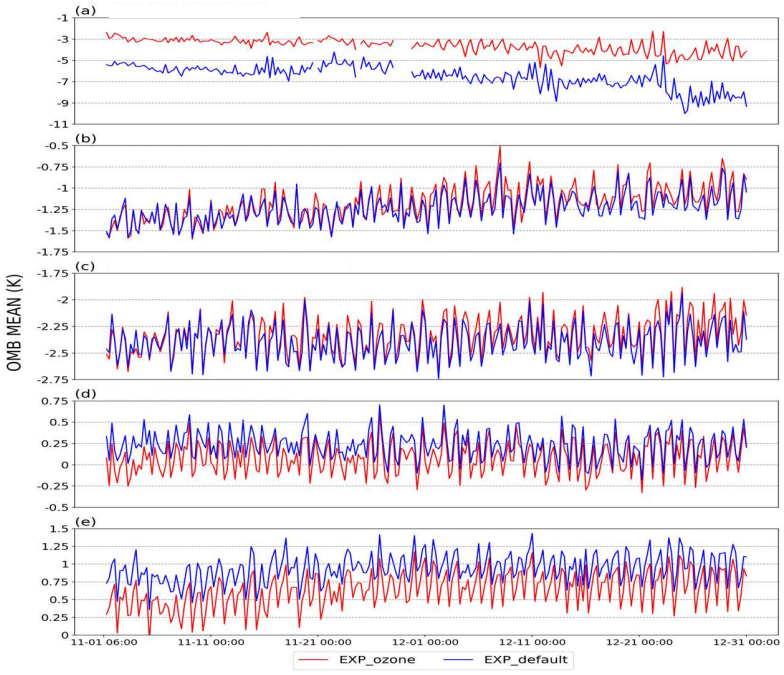


Fig. 5. Mean OMB values of EXP_default (blue) and EXP_ozone (red) in the Arctic for all assimilation moments from 0600UTC on 1 November to 0000UTC on December 31, 2023. (a) NOAA18-HIRS4 9.71 μm, (b) FY3C-MWHS-II 183.31 $\pm$ 1.0 GHz, (c) FY3D-MWHS-II 183.31 $\pm$ 1.0 GHz, (d) SNPP-ATMS 183.31 $\pm$ 1.0 GHz, and (e) MetOpC-MHS 183.31 $\pm$ 1.0 GHz band.

of predictors during assimilation. These predictors, which include model variables, satellite observation parameters, and their linear combinations, are dynamically optimized at each assimilation cycle [24]. As the use of spatiotemporally varying ozone reduces the brightness temperature simulation error, the required bias correction will be reduced accordingly.

As shown in Fig. 7, we counted the temporal variations of coefficients of the constant offset P0 and air mass predictors (P1: 1000–300 hPa atmospheric thickness, P2: 200–50 hPa

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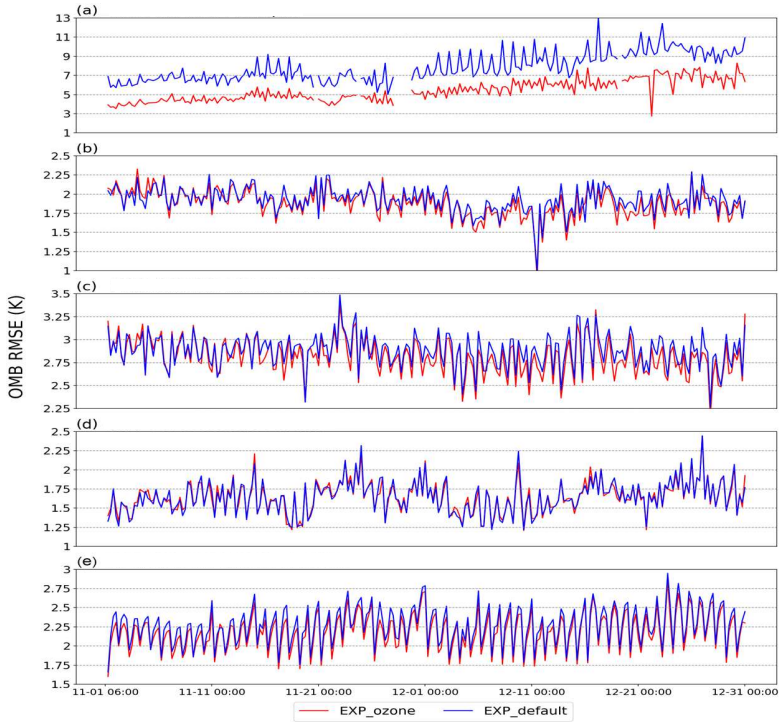


Fig. 6. RMSE of OMB values of EXP_default (blue) and EXP_ozone (red) in the Arctic for all assimilation moments from 0600UTC on 1 November to 0000UTC on December 31, 2023. (a) NOAA18-HIRS4 9.71 μm, (b) FY3C-MWHS-II 183.31 $\pm$ 1.0 GHz, (c) FY3D-MWHS-II 183.31 $\pm$ 1.0 GHz, (d) SNPP-ATMS 183.31 $\pm$ 1.0 GHz, and (e) MetOpC-MHS 183.31 $\pm$ 1.0 GHz band.

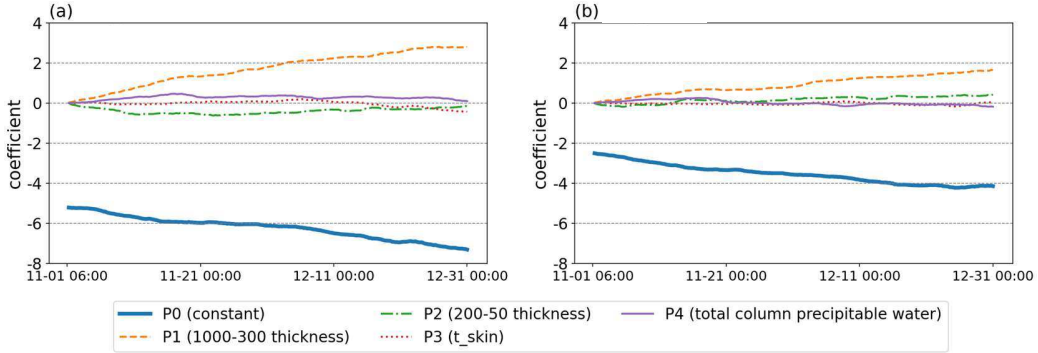


Fig. 7. Temporal variation of the coefficients of the VarBC predictors from P0 to P4 of NOAA18-HIRS4 9.71 μm band. (a) EXP_default and (b) EXP_ozone.

atmospheric thickness, P3: surface temperature, and P4: total column precipitable water) for the HIRS4 9.71 μm band during the assimilation cycle. Both the stabilized coefficients of P0 and air mass predictors, using spatiotemporally varying ozone during the assimilation cycle, have smaller values relative to EXP_default. Furthermore, these coefficients can stabilize more quickly with time. For the microwave 183.31 $\pm$ 1.0 GHz band, the difference between the VarBC coefficients of two experiments is not significant due to the minimal improvement of the brightness temperature simulation by the spatiotemporally varying ozone profiles and the bias itself (figure not shown).

Fig. 8 shows the frequency distributions of OMB, the innovation vector INV (OMB after bias correction), and OMA at the last assimilation moment. For the 9.71 μm band of NOAA18-HIRS4, not only does the use of spatiotemporally varying ozone reduces the error before bias correction, but also the frequency distribution of INV is much closer to a Gaussian normal distribution than that using the default profile. Although the microwave 183.31 $\pm$ 1.0 GHz bands are not as sensitive to ozone as the infrared band, their INV with spatiotemporally varying ozone are still closer to the Gaussian normal distribution. Both the mean and RMSE of OMB and INV also reduce to some extent, with the smaller errors of INV compared to OMB demonstrating the effectiveness of the bias correction. In addition, the successful assimilation of the satellite radiance data is evidenced by the closer alignment with the normal distribution of OMA. These results show that the use of spatiotemporally varying ozone not only improves the bias correction results but also reduces the cost of VarBC during assimilation.

B. Amount of Data Assimilated

Improved brightness temperature simulations and bias correction result in enhanced quality control pass rates. This

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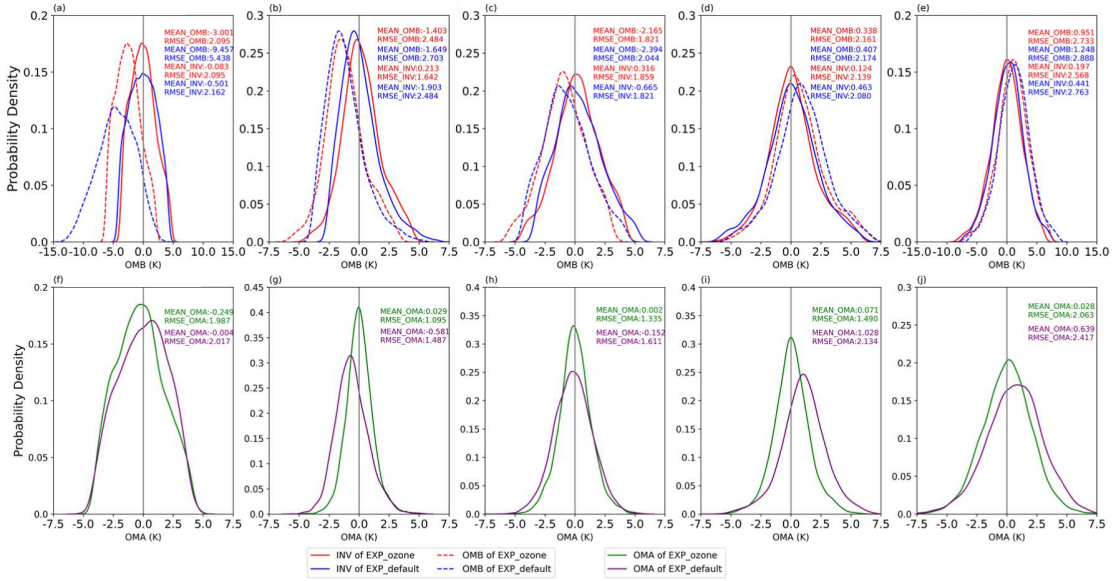


Fig. 8. INV frequency distribution of EXP_default (blue) and EXP_ozone (red) at the last assimilation moment. (a) NOAA18-HIRS4 9.71 μm band, (b) FY3C-MWHS-II 183.31 $\pm$ 1.0 GHz band, (c) FY3D-MWHS-II 183.31 $\pm$ 1.0 GHz band, (d) SNPP-ATMS 183.31 $\pm$ 1.0 GHz band, and (e) MetOpC-MHS 183.31 $\pm$ 1.0 GHz band.

TABLE I

MEAN NUMBER OF ASSIMILATED OBSERVATIONS OF THE TWO ASSIMILATION EXPERIMENTS AT EACH ANALYSIS MOMENT FROM 0600 UTC ON 1 NOVEMBER TO 0000 UTC ON DECEMBER 31, 2023

	NOAA18-HIRS4 9.71μm	FY3C/D-MWHS-II 183.31±1.0GHz	SNPP-ATMS 183.31±1.0GHz	METOPC-MHS 183.31±1.0GHz
EXP_default	3452	10060	8289	13421
EXP_ozone	3580	10098	8312	13435

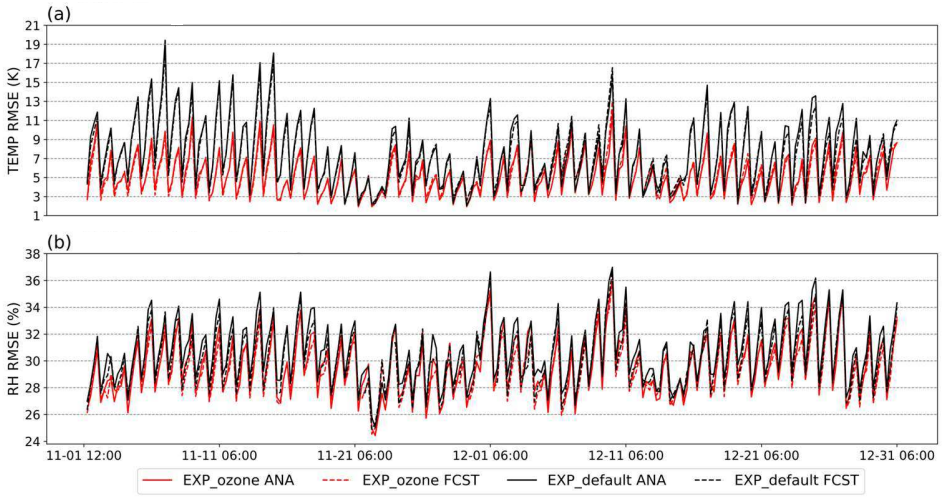


Fig. 9. RMSE time series of the analyses (solid line) and the corresponding forecasts (dashed line) for EXP_default (black) and EXP_ozone (red) in all assimilation moments from 2023-11-01-06UTC to 2023-12-31-00UTC. (a) 10 hPa temperature and (b) 400 hPa relative humidity.

is displayed by the statistics of the number of assimilated observations averaged across the analysis moments from 0600 UTC November 1 to 0000 UTC December 31, 2023, in Table I. For the 9.71 μm band of NOAA18-HIRS4, the average number of observations assimilated increases from 3452 to 3580 with an increase of 3.57% in observation usage. For the 183.31 $\pm$ 1.0 GHz bands of three microwave sensors, the observation usage increased by 0.38%, 0.28%, and 0.1%, respectively.

C. Assimilations and Forecasts

In order to quantitatively evaluate the effect of ozone profile selection on the assimilations and forecasts, the RMSE

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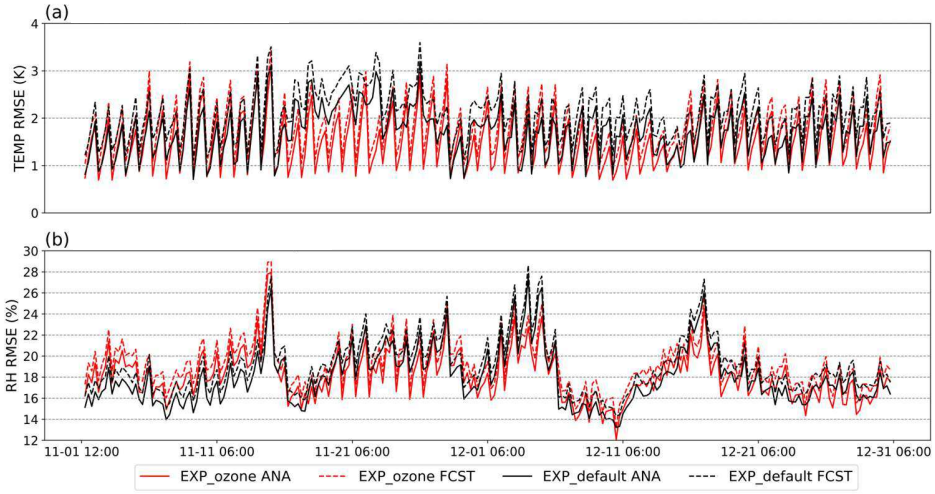


Fig. 10. RMSE time series of the analyses (solid line) and the corresponding forecasts (dashed line) for EXP_default (black) and EXP_ozone (red) in all assimilation moments from 2023-11-01-06UTC to 2023-12-31-00UTC. (a) 200 hPa temperature and (b) 200 hPa relative humidity.

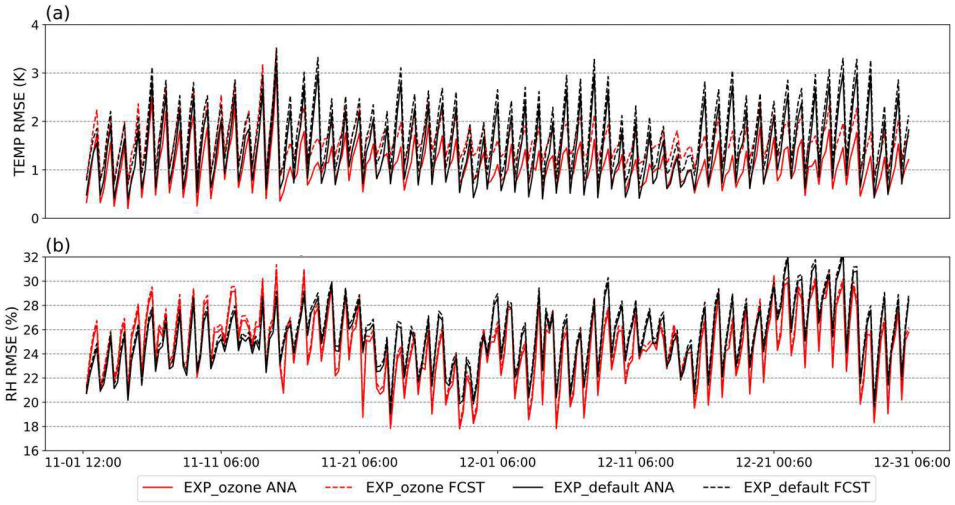


Fig. 11. RMSE time series of the analyses (solid line) and the corresponding forecasts (dashed line) for EXP_default (black) and EXP_ozone (red) in all assimilation moments from 2023-11-01-06UTC to 2023-12-31-00UTC. (a) 500 hPa temperature and (b) 500 hPa relative humidity.

of temperature, relative humidity, water vapor mixing ratio, and geopotential height are calculated for EXP_default and EXP_ozone, using the FNL reanalysis as validation. Fig. 9 firstly presents the RMSE time series of stratosphere temperature (10 hPa) and mid-troposphere humidity (400 hPa) for all the analysis and forecast fields (the 6-h forecast from the previous analysis time, which serves as the background field at the current time), where the selected height corresponds to the sensors' weight function. For the temperature analysis at 10 hPa, EXP_ozone consistently outperforms EXP_default at all assimilation moments, and the analysis errors are significantly reduced by using the spatiotemporally varying ozone during November 5–15 and December 15–25. For the relative humidity analysis at 400 hPa, EXP_ozone also outperforms EXP_default, with error reductions ranging from 0.5% to 1%. The characteristics of the forecasts for both temperature and relative humidity are similar to the analysis, but the error reduction of the forecasts using spatiotemporally varying ozone is relatively smaller.

This study additionally examined the RMSE of temperature and relative humidity at two representative pressure levels, 200 (Fig. 10) and 500 hPa (Fig. 11). The results indicate that the positive impact of using the spatiotemporally varying ozone is not only reflected at the altitudes primarily observed by the instruments but also propagates to other levels.

The vertical RMSE profiles of the four variables (temperature, relative humidity, water vapor mixing ratio, and geopotential height) averaged over the analysis and forecast fields for the two-month assimilation experiment are given in Fig. 12. Considering the actual atmospheric ozone distribution and the range of the instrumental observational heights, only the results above 500 hPa are shown. The RMSE of the temperature analysis of EXP_ozone most significantly decreases relative to EXP_default in the 20–5 hPa range. The RMSE of the relative humidity analysis of EXP_ozone is smaller than that of EXP_default in both the altitude ranges from 20 to 5 hPa and from 500 to 100 hPa, with a maximum error reduction of 0.8% at 400 hPa. Because of the small order of magnitude

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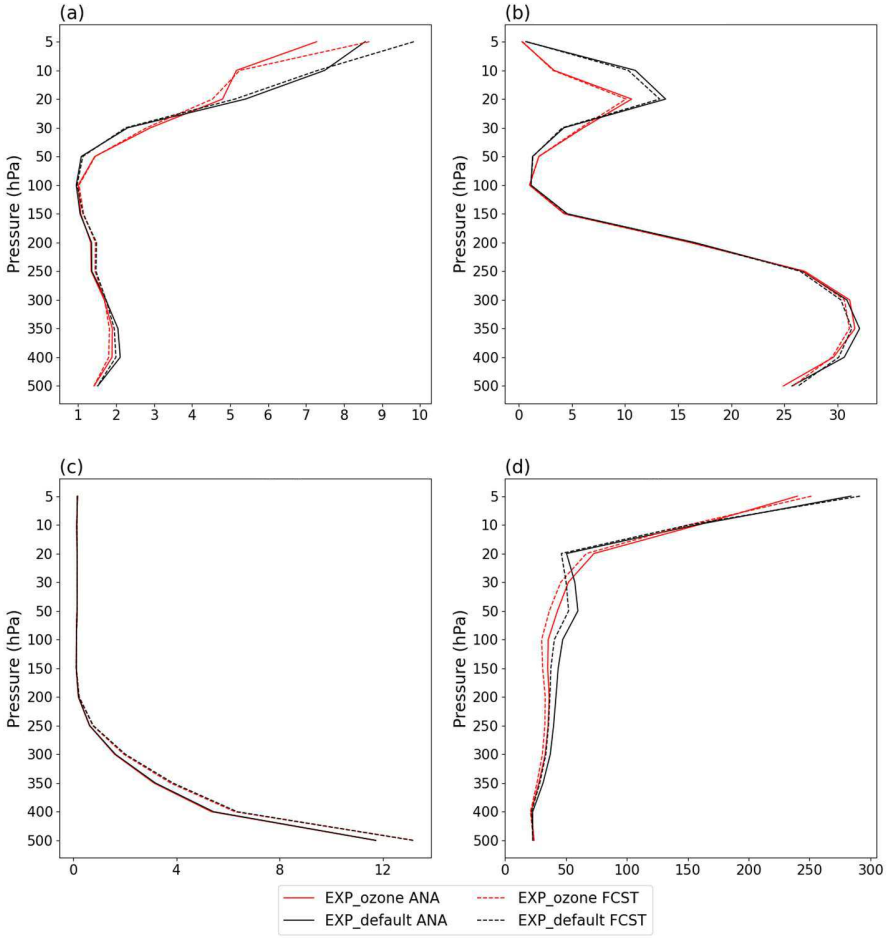


Fig. 12. Vertical profiles of RMSE of EXP_default (black) and EXP_ozone (red) for each variable in analysis fields (solid line) and corresponding forecast fields (dashed line) from 2023-11-01-06 UTC to 2023-12-31-00 UTC. (a) Temperature (K), (b) relative humidity (%), (c) water-vapor mixing ratio (10^{-5} kg/kg), and (d) geopotential height (gpm).

of the water vapor mixing ratio itself above 400 hPa, the difference in the water vapor mixing ratio analysis between the two experiments is not significant, with EXP_ozone slightly outperforming EXP_default. The improvement in geopotential height analysis of EXP_ozone relative to EXP_default, on the other hand, increases with the height, with an error reduction of 12 gpm at 5 hPa. The overall vertical characteristics of the RMSE in forecast fields are basically the same as those in the analysis. The forecast errors of EXP_ozone in stratospheric temperature and mid-troposphere relative humidity are also reduced relative to EXP_default. The water vapor mixing ratio and geopotential heights forecasts of EXP_ozone are also closer to the validation data. Overall, the use of spatiotemporally varying ozone when assimilating both infrared and microwave specific bands provides significant improvements in the analyses and forecasts of temperature, relative humidity, and geopotential height.

It is notable that in the previous multiinstrument joint assimilation experiments, HIRS4 9.71 μm demonstrated strong sensitivity to ozone. Therefore, we assimilate this observation data individually here to analyze its impact alone.

The vertical RMSE profiles of temperature, relative humidity, water vapor mixing ratio, and geopotential height averaged over the analysis and forecast fields for the two-month experiment are given in Fig. 13. The differences between the analysis and forecast fields of the two experiments are primarily observed in the stratosphere. In the case of assimilating only HIRS4 observations, the RMSE of the analysis and forecast of temperature in EXP_ozone remains significantly lower than those in EXP_default above 30 hPa, showing a notable reduction. Within this altitude range, the error of relative humidity analysis or forecast in EXP_ozone also decreases markedly. In addition, the geopotential height analysis and forecast above 200 hPa of EXP_ozone is slightly better than that of EXP_default. When the aforementioned microwave observations are not assimilated, there is almost no difference in the analysis and forecasts of the two experiments in the troposphere.

It should be noted that the 9.71 μm channel is not a humidity-sounding channel. Combined with Fig. 13, the differences in relative humidity analysis between the two experiments are mainly caused by differences in temperature analysis, with the results from November 20 to December 1 showing particularly significant differences in Fig. 14. Additionally, although the relative humidity values at this altitude are generally low, both experiments exhibit large RMSE,

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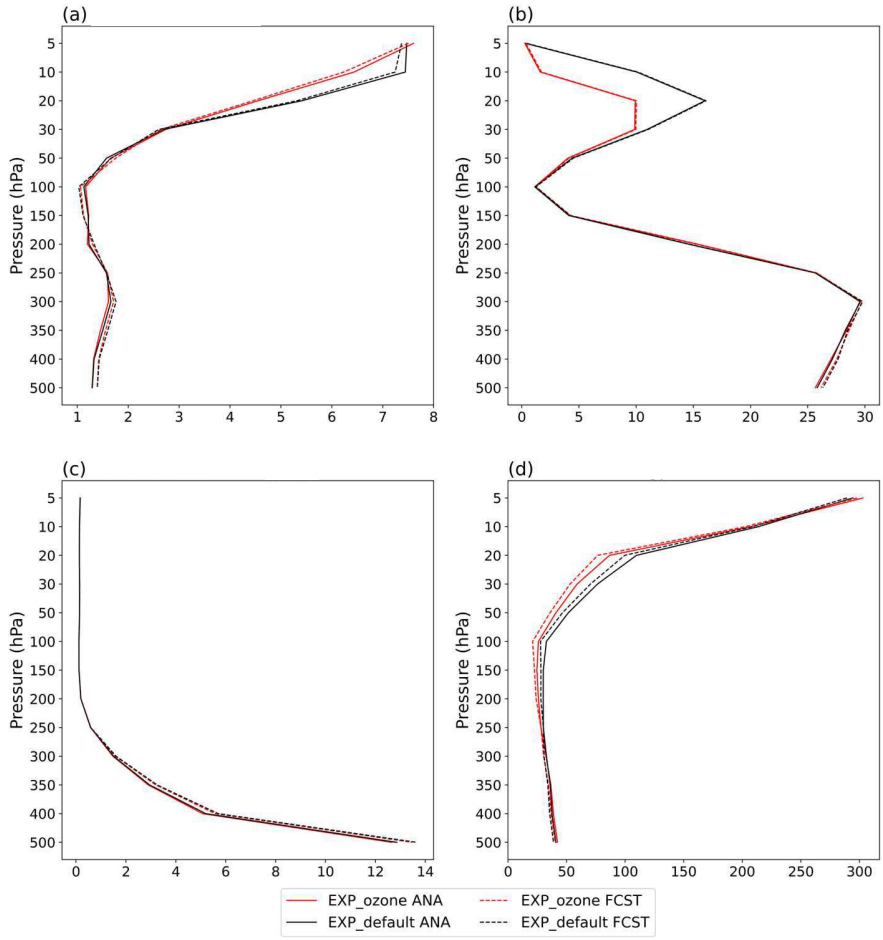


Fig. 13. Vertical profiles of RMSE of EXP_default (black) and EXP_ozone (red) for each variable in analysis fields assimilating HIRS4 9.71 μm observations only (solid line) and corresponding forecast fields (dashed line) from 2023-11-01-06 UTC to 2023-12-31-00 UTC. (a) Temperature (K), (b) relative humidity (%), (c) water-vapor mixing ratio (10^{-5} kg/kg), and (d) geopotential height (gpm).

which may still be related to current model errors in the upper atmosphere. This is an aspect that requires improvement in the future.

It is worth noting that the results of the observation operators depend on the accuracy of the forecast since the assimilation analysis requires the forecast at the previous moment as the background. The improvement of the analysis and forecast due to the use of spatiotemporally varying ozone at the first moment can be accordingly propagated to subsequent analyses and forecasts, while the VarBC predictors also emanates mainly from the background, which, in turn, improves the assimilation process. The mutual enhancement allows the spatiotemporally varying ozone to have a positive impact on assimilation analysis and forecast.

D. Diagnostic Analysis of Individual Case

Previous analyses have shown that the use of spatiotemporally varying ozone in place of the default profile is a positive contribution to brightness temperature simulation and bias correction, as well as to the cycle assimilations and forecasts. For a more in-depth understanding of the influence of spatiotemporal features of ozone on satellite radiance data assimilation, we conduct a detailed diagnostic analysis of an Arctic polar vortex event in mid-December 2023.

Fig. 15 shows the difference between the 100 hPa ERA5 ozone concentration and the default ozone profile at the first assimilation moment within the event (0600 UTC on December 11, 2023), the spatial distribution of OMB using the default ozone profile, and its difference with the results using spatiotemporal varying ozone (EXP_default-EXP_ozone). It can be seen that spatiotemporally varying ozone profiles effectively reduce simulation bias of infrared and microwave bands [see Fig. 15(c) and (e)], and this improvement is more pronounced in the regions where there are large differences in ozone concentrations between EXP_default and EXP_ozone, such as around 70°N and 120°–180°W Fig. 15(a). Morever, the use of spatiotemporally varying ozone profiles consequently reduces the amount of bias correction required, with HIRS4 9.71 μm and MHS 183.31 $\pm$ 1.0 GHz achieving about 2.5 and 0.3 K fewer corrections, respectively (figure not shown).

The 10 hPa temperature analysis increment and the 400 hPa relative humidity analysis increment for the two experiments at 0600 UTC December 11, 2023 are given in Fig. 16. In the region where default ozone profile shows significant underestimation, such as 70°N and 120°–180°W, the use of spatiotemporally varying ozone significantly reduces the negative bias in brightness temperature simulation and innovations [see Fig. 16(a)], resulting in a much smaller temperature

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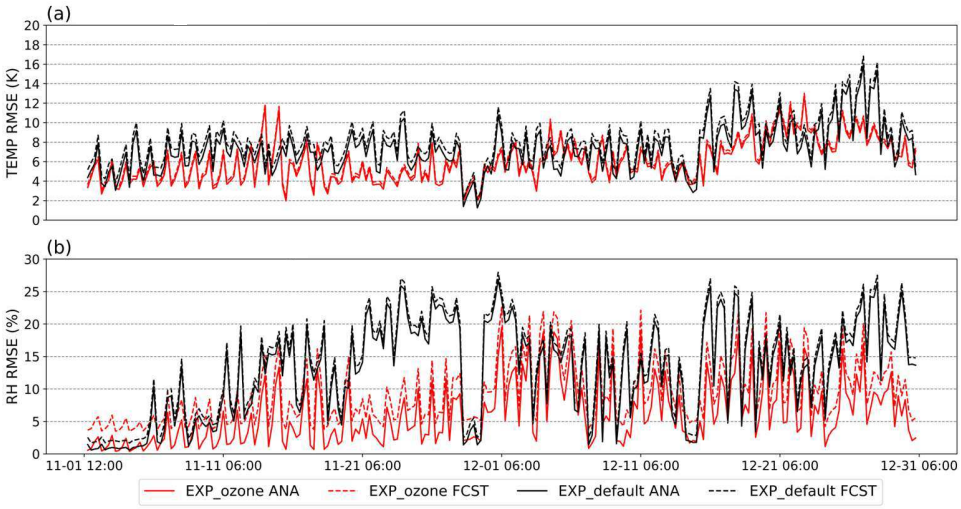


Fig. 14. RMSE time series of the analyses assimilating HIRS4 9.71 μm observations only (solid line) and the corresponding forecasts (dashed line) for EXP_default (black) and EXP_ozone (red) in all assimilation moments from 2023-11-01-06UTC to 2023-12-31-00UTC. (a) 10 hPa temperature and (b) 10 hPa relative humidity.

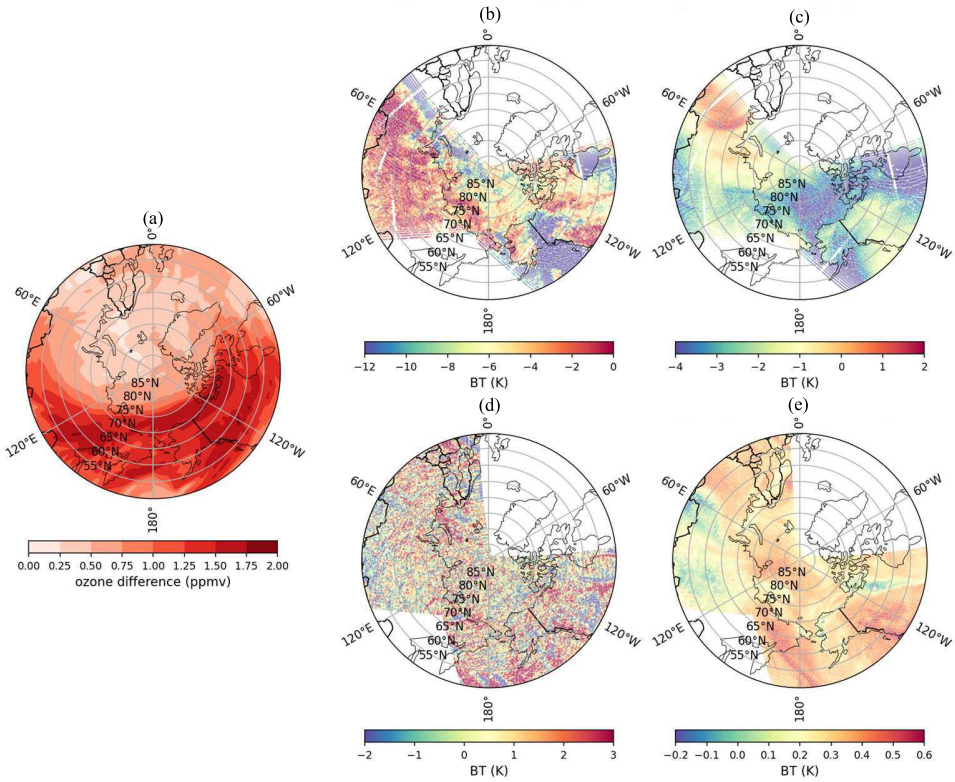


Fig. 15. (a) Differences between ERA5 ozone concentrations and default ozone profiles on 0600 UTC December 11, 2023 at 100 hPa, OMB spatial distributions using the default ozone profile and the differences with the results of that using spatiotemporally varying ozone (b) and (c) NOAA18-HIRS4 9.71 μm and (d) and (e) MetOpC-MHS 183.31 $\pm$ 1.0 GHz.

negative increments in EXP_ozone. Although the use of more realistic ozone profiles has a smaller improvement on the microwave brightness temperature simulations [see Fig. 16(e)], in the region where the two ozone concentrations differ significantly EXP_ozone still has a larger positive increment of relative humidity.

Fig. 17 displays the longitudinal cross sections (60°N to 90°N) along 30°E of the center of the polar vortex, showing the mean differences between ERA5 ozone concentration and RTTOV default ozone profile, as well as the temporal mean differences of temperature and relative humidity analysis between EXP_ozone and EXP_default from 0600 UTC 11 to 0000 UTC December 14, 2023. Fig. 17(a) reveals that the ERA5 ozone concentration used by EXP_ozone is lower than the RTTOV default profile used by EXP_default above 50 hPa, with the maximum difference occurring at 10 hPa. Between 250 and 50 hPa, the ERA5 ozone concentrations are higher than the RTTOV default profile. The peak height of the ozone Jacobian in the HIRS4 9.71 μm band is near 300 hPa, while the ERA5 ozone concentration is lower than the default

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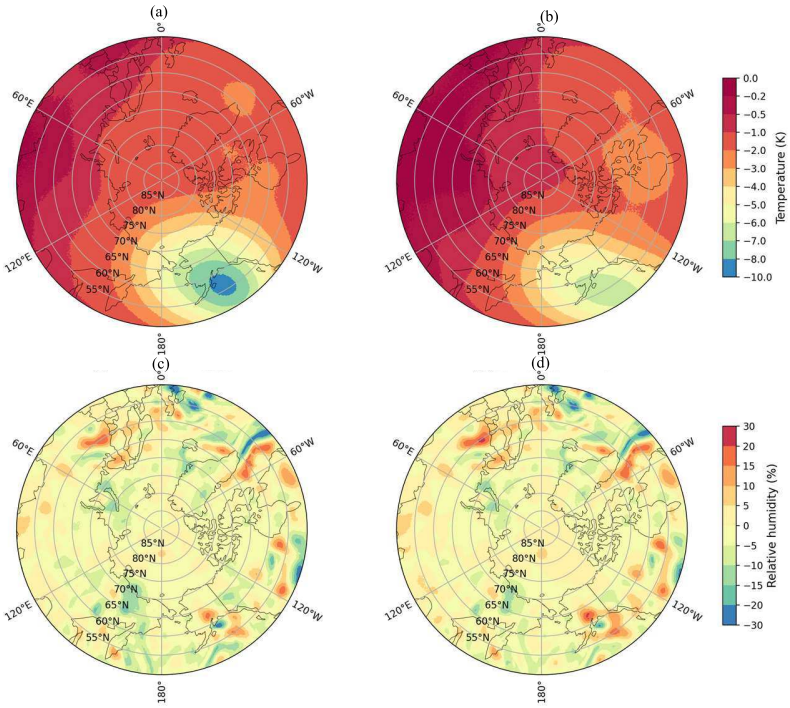


Fig. 16. 10 hPa temperature analysis increments for (a) EXP_default and (b) EXP_ozone, and 400 hPa relative humidity analysis increments for (c) EXP_default and (d) EXP_ozone, at 0600 UTC December 11, 2023.

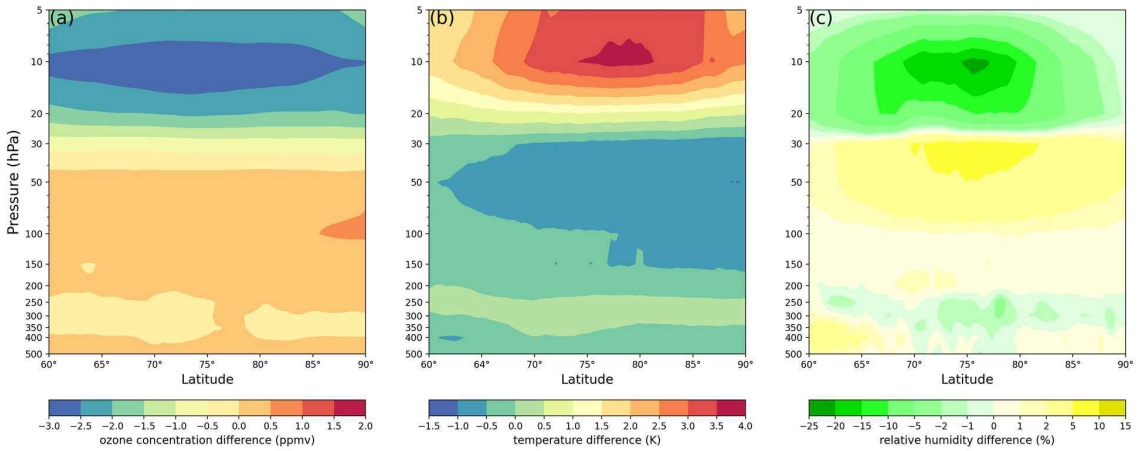


Fig. 17. Longitudinal profiles (60°N to 90°N) along 30°E at the center of the polar vortex for (a) mean differences between ERA5 ozone concentrations and RTTOV default ozone profiles, (b) mean difference of temperature analysis between EXP_ozone and EXP_default, and (c) mean difference of relative humidity analysis between EXP_ ozone and EXP_default from 0600 UTC 11 to 0000 UTC December 14, 2023.

profile. Since there is a strong negative correlation between radiance at 9.71 μm and ozone concentration, the use of the spatiotemporally varying ozone reduces the negative bias in brightness temperature simulation. As a result, temperature analysis of EXP_ozone is significantly higher than that of EXP_default at 20–10 hPa, which is the primary observation layer for this band [see Fig. 17(b)]. At constant specific humidity, this warming leads to lower relative humidity, explaining why the relative humidity analysis at this altitude is lower for EXP_ozone than for EXP_default. The radiance brightness temperature in the microwave 183.31 $\pm$ 1.0 GHz band has a negative correlation with the ozone concentration between 250 and 50 hPa. Consequently, the relative humidity analyses of the two experiments differ from 500 to 300 hPa, with EXP_ozone being drier than EXP_default [see Fig. 17(c)].

Validation data is introduced to compare the accuracy of the two sets of experimental analysis results. Temperature and humidity profiles provided by radiosonde observations from stations 71 082 (82.493°N, 62.344°W) and 22 008 (68.990°N, 33.13°E) were selected to calculate absolute deviations. As shown in Fig. 18, the assimilation using spatiotemporally varying ozone produces the stratospheric temperatures and mid-tropospheric relative humidity in the analysis fields that exhibit better agreement with the observations.

The 24-h deterministic forecast was initialized using the assimilated analysis field from the final moment of the daily assimilation cycle. Fig. 19 presents the difference between the 6, 12, 18, and 24 h 10-hPa temperature forecasts and FNL reanalysis. At 10 hPa, FNL at 0600UTC on 12 December reveals a large cold region coinciding with the stratospheric

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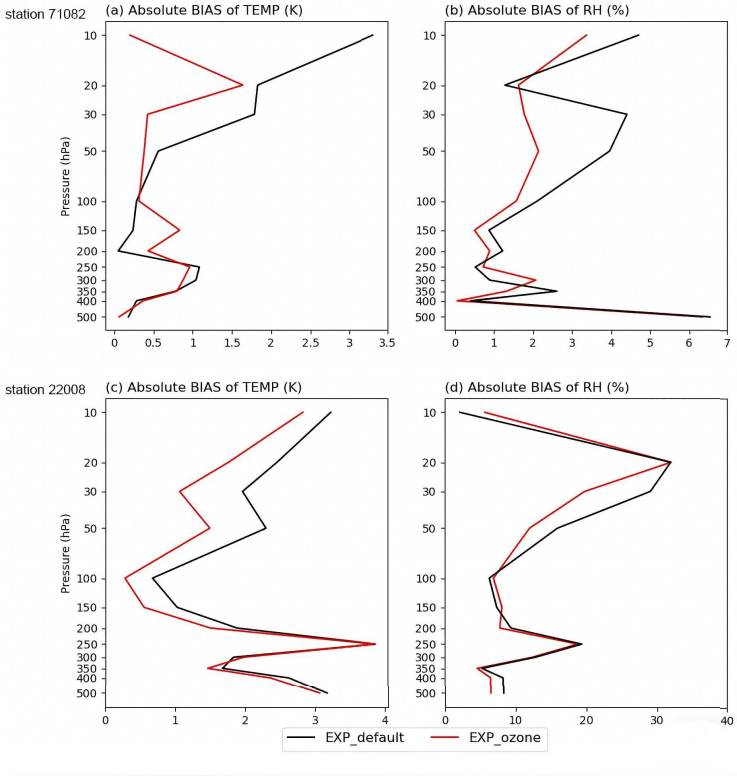


Fig. 18. Absolute deviation vertical profiles of EXP_default and EXP_ozone relative to radiosonde observations at stations (a) and (b) 71 082 and (c) and (d) 22 008 from 0600 UTC 11 to 0000 UTC December 14, 2023, (a) and (c) temperature and (b) and (d) relative humidity.

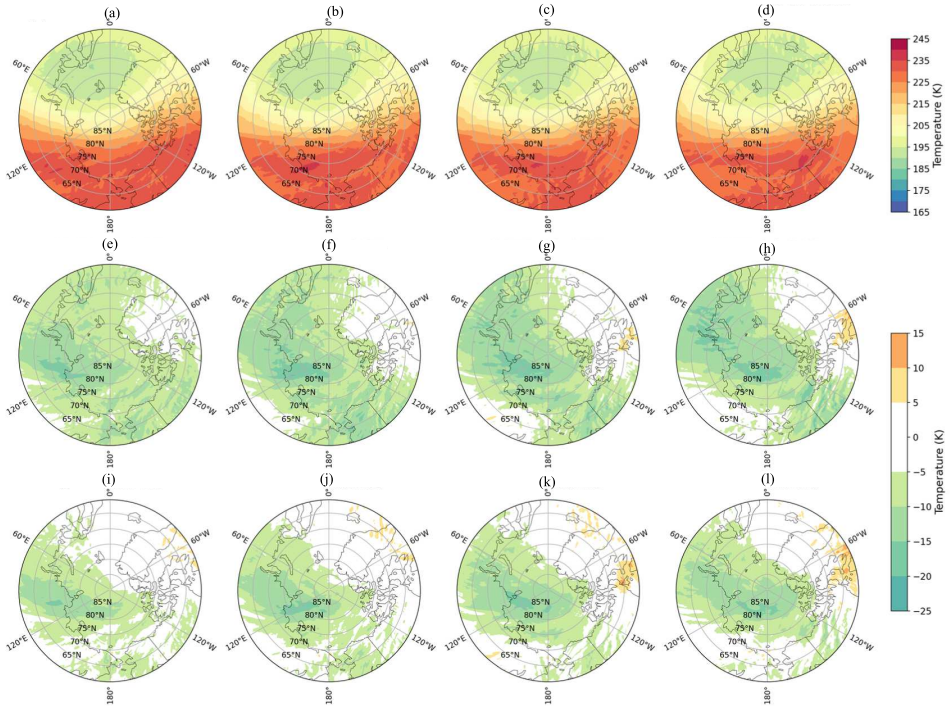


Fig. 19. 10 hPa temperature of FNL and its difference between EXP_default, EXP_ozone deterministic forecast results and FNL (filled colors, K). (a)~(d) FNL reanalyses by 6-h interval from 0600 UTC 11 to 0000 UTC December 14, 2023, (e)~(h) difference between EXP_default's 6, 12, 18, and 24 h forecasts with FNL, and (i)~(l) difference between EXP_ozone's 6, 12, 18, and 24 h forecasts with FNL.

polar vortex centered near 75°N, 30°E, where minimum temperatures approach 185 K. South of 75°N, a warmer zone extends from 70°W to 120°E. The EXP_default experiment exhibits significant cold biases in both regions, with temperature underestimations reaching 10 K at the vortex center and 10–20 K in the warmer southern region compared to FNL. EXP_ozone mitigates the negative temperature bias. During subsequent forecast times (12, 18, and 24 h), although temper-

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ature forecast errors increase in both experiments, EXP_ozone maintains smaller overall discrepancies with FNL despite a slight warm bias near 70°W. These results confirm that the optimization of the initial fields in the stratosphere and upper troposphere by the use of spatiotemporally varying ozone is also passed on to the subsequent forecasts, as compared to the use of the default ozone profile. EXP_ozone produces 10 hPa temperature forecasts at 6–24 h lead times that better match FNL, particularly in correcting the cold bias exhibited by EXP_default.

VI. Conclusion

While ozone has a certain effect on the observed radiance in specific infrared and microwave bands, current satellite radiance data assimilation often oversimplifies spatiotemporal variable, like ozone, as a single static profile. It reduces the accuracy of radiance simulation for the corresponding bands, and ultimately impacts assimilation effectiveness. In this study, a satellite radiance data assimilation scheme considering the spatiotemporal variability of ozone is implemented, and a two-month continuous assimilation and forecast experiment is conducted in the Arctic with infrared and microwave observations of NOAA18-HIRS4, FY3C/D-MWHS-II, SNPP-ATMS, and MetOpC-MHS. The main conclusions are as follows:

1) Ozone concentration in the Arctic has significant spatial and temporal variability, and the default ozone profile of RTTOV shows significant discrepancies against spatiotemporally varying ozone concentration at different moments and satellite observation locations.
2) The use of spatiotemporally varying ozone profiles effectively reduces the brightness temperature simulation bias in both HIRS4 9.71 μm and microwave 183 $\pm$ 1.0 GHz bands of MWHS-II, ATMS, and MHS. For infrared observations, the implementation decreases OMB mean and RMSE by up to 5 K compared to using the default profile. Although microwave 183.31 $\pm$ 1.0 GHz bands are less sensitive to ozone than infrared specific bands, the use of spatiotemporally varying ozone profiles still improves the simulation accuracy and centers the overall bias distribution closer to zero.
3) The use of spatiotemporally varying ozone profiles not only accelerates the stabilization of the VarBC coefficient, but also improves the results of the correction. This enhancement increases the number of observations that pass the quality control into the assimilation and reduces the analysis errors of physical variables. The 9.71 μm and 183 $\pm$ 1.0 GHz bands observations have a significant influence on the analysis of the 10-hPa temperature and 400-hPa relative humidity, respectively.
4) Experiments using static and dynamic ozone profiles (EXP_default and EXP_ozone) produce great discrepancies in analyses and forecasts. The analyses and forecasts using the spatiotemporally varying ozone are much closer to the validation data, effectively improving the low temperature forecasts and consistently mitigating the 10 hPa cold bias throughout the 24-h forecast period, though the improvement gradually diminishes with longer forecast time.

Although spatiotemporally varying ozone finally resulted in a more reasonable analysis and forecast of the Arctic atmosphere, the persistent simulation errors, particularly at high altitudes, potentially suggest the model limitations in simulating the upper Arctic atmosphere [25]. Regarding observational capabilities, the existing microwave observation frequency band is up to about 183 GHz, and in the future, ICI on the Metop-SG series satellites and the Chinese geostationary satellite microwave sounder have higher frequency bands [26], [27]. Greater ozone sensitivity of these advanced instruments will require more rigorous consideration of ozone's spatiotemporal characteristics in assimilation. If this method is to be used for future operational implementation, the main steps follow the manuscript and we could use real-time ozone products from global forecasting systems (e.g., ECMWF or NCEP) and near-real-time satellite-retrieved ozone products. It would also be feasible to integrate a chemical model forecast to drive the radiative model for assimilation.

In the context of current systems, the OMB in the infrared ozone-sensitive band is significantly affected by ozone, yet the current bias correction module does not account for its impact on systematic errors. Therefore, our next step is to start with infrared observations and incorporate real-time ozone data as an additional bias correction predictor into the VarBC module.

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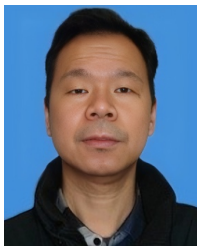


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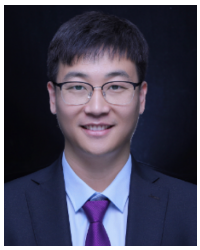
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